

1 3 0 2 3 1 0 2 1 2 3 1 0
 01 11 00 01 11 01 00 10 01 10 11 01 00
 10 11 01 11 01 10 10 11 10 10

10 11 01 11 01 10 10 11 10 10

N symbol. 2N $\rightarrow$ b./symbol = 2

$$N_{sym} \approx \frac{N}{2} \times 1 + \frac{N}{4} \times 2 + \frac{N}{8} \times 3 + \frac{N}{8} \times 3 = N \left[\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{3}{8} \right] = N \frac{9+4+3+3}{8} = N \frac{14}{8}$$

$$I(A \cap B) = \log \left(\frac{1}{P(A \cap B)} \right) = \log \left(\frac{1}{P(A) \cdot P(B)} \right) = \log \left(\frac{1}{P(A)} \cdot \frac{1}{P(B)} \right) = \log \left(\frac{1}{P(A)} \right) + \log \left(\frac{1}{P(B)} \right) = I(A) + I(B)$$

b./symbol = 1.75

$$P(A_1) \log \frac{1}{P(A_1)} + P(A_2) \log \frac{1}{P(A_2)} + \dots + P(A_n) \log \frac{1}{P(A_n)}$$

A.1 $\lim_{x \rightarrow 0} x \log \frac{1}{x}$

A=2
 $\lambda_n = 2^{-n}$

$$2^{-n} \log_2 \frac{1}{2^{-n}} = 2^{-n} \log_2 2^n = n 2^{-n} \rightarrow 0$$

0	$\frac{1}{4}$	$\frac{1}{2}$
1	$\frac{1}{4}$	$\frac{1}{4}$
2	$\frac{1}{4}$	$\frac{1}{8}$
3	$\frac{1}{4}$	$\frac{1}{8}$

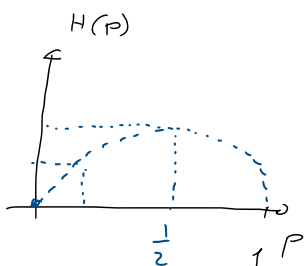
$$4 \left\{ \frac{1}{4} \log_2 \frac{1}{1/4} \right\} = \log_2 (4) = 2 \text{ b./c}$$

$$\frac{1}{2} \log \frac{1}{1/2} + \frac{1}{4} \log \frac{1}{1/4} + \frac{1}{8} \log \frac{1}{1/8} + \frac{1}{8} \log \frac{1}{1/8}$$

$$= \frac{1}{2} \log 2 + \frac{1}{4} \log 4 + \frac{2}{8} \log 8$$

$$= \frac{1}{2} \times 1 + \frac{1}{4} \times 2 + \frac{2}{8} \times 3 = \frac{1}{2} + \frac{1}{2} + \frac{6}{8} = 1 + \frac{6}{8} = \frac{14}{8}$$

$$\frac{1}{2} \log \frac{1}{1/2} + \frac{1}{2} \log \frac{1}{1/2} = \log 2 = 1$$



$$p = \frac{1}{4}$$

$$\frac{1}{4} \log \frac{1}{1/4} + \frac{3}{4} \log \frac{1}{3/4} = -\frac{1}{4} \log 4 + \frac{3}{4} (\log 4) - \frac{3}{4} \log (3)$$

$$= \frac{1}{2} \log (4) - \frac{3}{4} \log (3)$$

N b.i.2

N H(p) b.i.2

Gennerio 2026

10 "1"

6 "2"

4 "3"

$X_1 \backslash X_2$	1	2	3	
1	$\frac{10}{20} \cdot \frac{9}{19}$	$\frac{10}{20} \cdot \frac{6}{19} = \frac{6}{19}$	$\frac{10}{20} \cdot \frac{4}{19}$	$\frac{10}{20} = \frac{5}{10}$
2	$\frac{6}{20} \cdot \frac{10}{19}$	$\frac{6}{20} \cdot \frac{5}{19}$	$\frac{6}{20} \cdot \frac{4}{19}$	$\frac{6}{20} = \frac{3}{10}$
3	$\frac{4}{20} \cdot \frac{10}{19}$	$\frac{4}{20} \cdot \frac{6}{19}$	$\frac{4}{20} \cdot \frac{3}{19}$	$\frac{4}{20} = \frac{2}{10}$
	$\frac{10}{20}$	$\frac{6}{20}$	$\frac{4}{20}$	

$$P[X_1 = i] = \sum_k P[X_1 = i | X_2 = k] P[X_2 = k] = \sum_k P[X_1 = i, X_2 = k]$$

$$P[X_2 = 1] = \sum_k P[X_1 = k, X_2 = 1]$$

$$\frac{10}{20} \cdot \frac{9}{19} + \frac{6}{20} \cdot \frac{10}{19} + \frac{4}{20} \cdot \frac{10}{19} = \frac{1}{19} \left[\frac{9}{2} + \frac{3}{1} + \frac{2}{1} \right] = \frac{1}{19} \frac{9+10}{2} = \frac{1}{2}$$

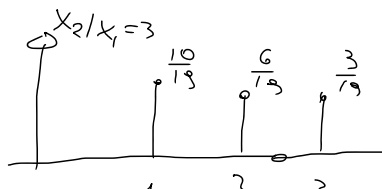
$$\frac{10}{20} \cdot \frac{6}{19} + \frac{6}{20} \cdot \frac{5}{19} + \frac{4}{20} \cdot \frac{6}{19} = \frac{1}{19} \left[\frac{3}{1} + \frac{3}{2} + \frac{6}{5} \right] = \frac{1}{19} \frac{30+15+12}{10} = \frac{57}{19} \frac{1}{10} = \frac{3}{10}$$

$$\frac{36}{20} = \frac{18}{10} = \frac{9}{5}$$

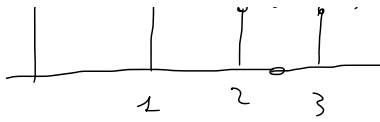
$$E[X_1] = 1 \times \frac{1}{2} + 2 \times \frac{3}{10} + 3 \times \frac{2}{10} = \frac{5+6+6}{10} = \frac{17}{10}$$

$$E[X_2 | X_1 = 3]$$

$$P[X_2 = i | X_1 = 3] = \frac{P[X_1 = 3, X_2 = i]}{P[X_1 = 3]} = \frac{P[X_1 = 3, X_2 = i]}{\sum_{k=1}^3 P[X_1 = 3, X_2 = k]}$$



$$1 \times \frac{10}{19} + 2 \times \frac{6}{19} + 3 \times \frac{3}{19} = \frac{31}{19}$$



$$1 + \frac{10}{19} + 2 + \frac{6}{19} + 3 + \frac{3}{19} = \frac{31}{19}$$

$$P_{X_1}(k) = e^{-\lambda_1} \frac{\lambda_1^k}{k!} \quad k=0,1,2,\dots \quad Y = X_1 + X_2 \quad Y|_{X_1=m} = m + X_2$$

$$P_{X_2}(k) = e^{-\lambda_2} \frac{\lambda_2^k}{k!} \quad k=0,1,2,\dots$$

$$Y = m_1 + m_2$$

$$P_Y(k) = P[Y=k] = \sum_{m=0}^{+\infty} P[Y=k | X_1=m] \cdot P[X_1=m] = \sum_{m=0}^{+\infty} P[X_2=k-m] \quad \text{perché } m \leq k \text{ se } m > k$$

$$P[Y=k] = \sum_{m_1=0}^{+\infty} \sum_{m_2=0}^{+\infty} P[Y=k | X_1=m_1, X_2=m_2] P[X_1=m_1, X_2=m_2]$$

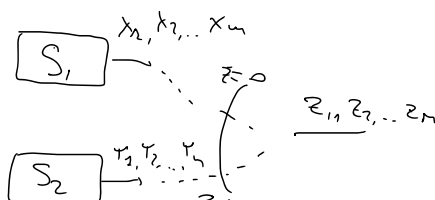
$$P[m_1+m_2=k] = \sum_{m_1=0}^{+\infty} \sum_{m_2=0}^{+\infty} \delta[k-m_1-m_2] P[X_1=m_1] P[X_2=m_2]$$

$$= \sum_{m_1=0}^{+\infty} \sum_{m_2=0}^{+\infty} \delta[k-m_1-m_2] P[X_1=m_1] P[X_2=m_2]$$

$$\sum_{m=0}^k e^{-\lambda_1} \frac{\lambda_1^m}{m!} e^{-\lambda_2} \frac{\lambda_2^{k-m}}{(k-m)!} = \frac{e^{-(\lambda_1+\lambda_2)}}{k!} \sum_{m=0}^k \frac{k!}{m!(k-m)!} \lambda_1^m \lambda_2^{k-m}$$

$$= \frac{e^{-(\lambda_1+\lambda_2)}}{k!} \sum_{m=0}^k \binom{k}{m} \lambda_1^m \lambda_2^{k-m} = \frac{e^{-(\lambda_1+\lambda_2)}}{k!} (\lambda_1 + \lambda_2)^k \quad \lambda_s = \lambda_1 + \lambda_2$$

$$P[Y=k] = \frac{e^{-\lambda_s}}{k!} \lambda_s^k$$



$$X_i \in \{0,1\} \quad P[X_i=0] = \frac{1}{3} \quad \dots \rightarrow P[X_i=1] = \frac{2}{3}$$

$$Y_i \in \{2,3\} \quad P[Y_i=2] = \frac{1}{4} \quad \dots \rightarrow P[Y_i=3] = \frac{3}{4}$$

$$P[\bar{E}=0] = 1$$

$$Z = X_i + Y_i$$

$$S_2 \rightarrow \begin{matrix} 1, 2, \dots, 5 \\ 2 \end{matrix}$$

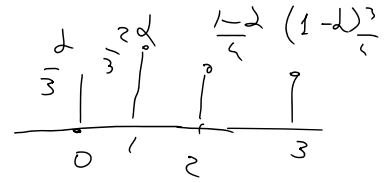
$$P[\bar{\epsilon}=0] = 1$$

$$Z_i = (1-\epsilon) X_i + \epsilon \tilde{Y}_i$$

$$Z_i / \epsilon=0 = X_i \quad \forall i$$

$$Z_i / \epsilon=1 = \tilde{Y}_i \quad \forall i$$

$$\begin{aligned} P[Z_1=0] &= P[\bar{\epsilon}=0] P[Z_1=0/\bar{\epsilon}=0] + P[\bar{\epsilon}=1] P[Z_1=0/\bar{\epsilon}=1] \\ &= 1 \cdot P[X_1=0] + (1-1) \cdot P[\tilde{Y}_1=0] = \frac{1}{3} \end{aligned}$$



$$\begin{aligned} P[Z_1=0, Z_2=0] &= P[\bar{\epsilon}=0] P[Z_1=0, Z_2=0/\bar{\epsilon}=0] + P[\bar{\epsilon}=1] P[Z_1=0, Z_2=0/\bar{\epsilon}=1] \\ &= 1 \cdot P[X_1=0, X_2=0] \\ &= 1 \cdot P[X_1=0] P[X_2=0] \\ &= 1 \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9} \end{aligned}$$

$$P[Z_1=0] P[Z_2=0] = \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$$

Giugno 2025

$$X \sim N(0, 1)$$

$$Y \sim N(1, 4)$$

$$\rho_{XY} = -\frac{1}{2}$$

$$Z = X - \frac{Y}{2}$$

1) PDF di Z

$$2) U = \begin{cases} 1 & \text{se } Z > \eta \\ -1 & \text{se } Z \leq \eta \end{cases}$$

PDF di U risp. di η

$$3) \eta: P[U=1] = P[U=-1]$$

Settembre 2025

$$Z \sim N(-1, 9)$$

$$Z = 3Y - 2X$$

$$\rho_{YZ} = \rho$$

1) PDF di Z risp. ρ

2) media e var. di U